

Production validation

Branch `claude/nac-ai-hvac-designer-9oda35`

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READY FOR NAC TO USE ON REAL CUSTOMER QUOTES?

NO

Not yet — and **not because of the code**. Every one of the five things standing in the way is a step on the live system that this environment cannot reach or perform: the database, the security policies, your material rates, and the ServiceM8 key. They are listed in order at the foot of this page. Nothing on that list needs a developer.

UNIT TESTS 388 24 files, all passing	BROWSER TESTS 16 suites, all passing	DEAD BUTTONS 0 of 97 wired	DEFECTS FIXED 10 4 of them mine
BLOCKERS LEFT 5 all yours to run			

Read this before the rest. Every persistence, security and integration result below was produced against a **stub**, because the network policy in this environment blocks `icnznjhwybryizbdqrgx.supabase.co` and `api.servicem8.com`. Those are marked **NOT TESTABLE** and no amount of work here will change that. A stub proves the application asks the right questions and reacts correctly to the answers. It proves nothing about your database.

01 The eight priorities

Three words only, as you asked: tested and passed, tested and failed, or not testable. Nothing here is called done on the strength of a stub.

P	WHAT WAS ASKED FOR	STATUS	WHAT THAT RESTS ON
P1	Live database validation Create, save, refresh, reload, edit, save, close, reopen — designs, quotes, settings, rooms, equipment, materials, revisions.	NOT TESTABLE	Supabase is unreachable from here. <code>/db-self</code> runs the whole round trip from your browser against the project and reports the HTTP status and database status at every step.
P2	Placeholder material pricing No placeholder may be treated as a verified supplier cost.	TESTED – PASS	52 material lines: 27 real, 25 placeholder, 0 unpriced line stops the quote outright; placeholder itemised confirmation naming the count, the dollar amount. The 25 rates are yours to set.
P3	Authentication and security No customer, pricing or design information to an unauthenticated visitor. Customer pages keep working.	TESTED – PASS	The gate holds on all three internal pages, no data is made before sign-in, and the staff token reaches the request. The RLS half is NOT TESTABLE until <code>designer/rls.sql</code> is applied — until then the page source still reads everything.
P4	Static pressure data Missing data must never produce a false pass.	TESTED – PASS	Three states, not two. 35 of 120 models have no ESP on file; the other 85 raise a blocking critical STATIC PRESSURE CHECK NOT COMPLETED going quiet. Nothing was invented to fill a gap.
P5	Responsive dialogs No native prompt or confirm. Works at iPad landscape.	TESTED – PASS	All 26 replaced. 48px buttons, 52px rows, 16px margins, does not zoom. Verified at 1180×820 and 390×844. test fails the run if any native dialog appears at all.
P6	Real PDF export Customer PDF must not expose supplier costs, margins or internal notes.	TESTED – PASS	Both documents generated from a real costed design read back with pdf.js: page size, no line crossing and no dollar figure anywhere in the customer PDF. One content model, two renderers, so they cannot be compared.
P7	Customer and job records Customer → job → design → quote. Do not destroy existing customer information.	TESTED – PASS	The whole chain written and read back across several scenarios. A merge fills blanks and never overwrites. live migration is NOT TESTABLE — <code>/crm-migrate.html</code> shows you the dry run first and nothing until you press APPLY.
P8	ServiceM8 Never show JOB CREATED unless ServiceM8 confirms it.	NOT TESTABLE	The API is blocked here and there is no key in the test environment. Three false success paths were found by reading the code, and the handler is tested against answers a real API gives — but it has never spoken to ServiceM8. <code>/api/servicem8-selftest</code> settles the deployed app.

02 Defects found and fixed

Ten, in the order they were found. Four were mine, introduced during this work — they are marked, because a report that only lists other people's mistakes is not worth reading.

The quote tool showed a blank page to anyone not signed in

MINE — P3

The sign-in gate hid the whole document, and the sign-in form lives inside that document. Opening `index.html` or `admin.html` gave a blank page — no form, no message, no way in. It was live on the branch for four commits. The worst defect in this round, and it was found by chasing a page health check that reported "0 characters" rather than writing it off.

A static pressure check that never happened looked like one that passed

EXISTING

The engine never invented a figure, but nothing said the comparison had not been made and nothing stopped approval. For 85 of 120 models that is a false pass.

A three-phase unit could be quoted with no mention of the supply

EXISTING

Found by running your brochure plan. On a single-phase house that is a supply upgrade discovered on installation day and never quoted. The Equipment tab now asks what the site has; unconfirmed ranks single-phase units first, and a site recorded as single-phase makes it critical.

The PDF font width table was misread, so text would overflow the page

MINE — P6

Parsed on a four-character stride while `@` is 1015 — four digits — so every width after it was read from the wrong offset. Customer documents would have had text running off the edge. There is now a test asserting the published Helvetica widths and that no wrapped line can exceed its column.

ServiceM8 answering 200 with no record id was reported as a created job

EXISTING

That is a job nobody can find. It is a failure now, with what ServiceM8 actually said. The job is also read back afterwards, which is both the confirmation and where the job number comes from.

When a ServiceM8 job failed to create, nobody found out

EXISTING

`sign.html` fired the request and dropped the result. Not the customer's problem — but it was not reaching you either. The outcome is written onto the quote row server-side now, where a closed tab cannot lose it.

Five browser tests were hanging forever, silently

MINE – P3

The sign-in gate broke them and they simply never finished, which is exactly what a real user would have experienced. Fixed, and `auth-gate.mjs` now checks the gate is on screen rather than merely present in the DOM — present-but-invisible was the whole of the blank page defect above, and the old assertion could not tell the difference.

CUP'D was counted as conditioned floor area

EXISTING

It is a cupboard. On your brochure plan it carries no size so it changed nothing, but on a plan where it does it would have gone straight into the load. CUP'D, CUPB, CPD, CUPBOARD, broom and cloak are all excluded now.

A second design at the same site interrupted the estimator on every save

MINE – P7

"Ducted install" then "Add a second system" is one job described two ways. The record keeps the first wording and nobody is asked about it.

A customer's second property was reported as a data entry error

MINE – P7

The address on a design is the **site** address, not the customer's own. It belongs to the job, where it always did.

03 Your plan, read end to end

The brochure ground-floor sheet you sent: twenty labelled rooms, eleven with a printed size, and two conditioned rooms with no size at all.

READ CORRECTLY 11/11 printed sizes	KEPT OUT OF LOAD 52.13 m² excluded, garage included	CONDITIONED 160.25 m² once MEALS and STUDY measured	TWO METHODS 0.9% apart — 23.2 vs 23.45 kW
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MEALS and STUDY have no size printed and it did not guess. They come through with no area and low confidence, get no airflow and no outlet, and the design will not approve until you deal with them. The garage — 41.4 m², the biggest room on the page and worth about 6 kW if read in — stays out, with the laundry, bath, ensuite, WC, pantry, porch and alfresco.

It then flagged what an estimator would: past a single unit, one return cannot carry 1199 L/s, eleven zones exceeds the controller's ten, and a material line with no cost blocks the quote. No duct above 400 mm anywhere. At 23.45 kW there is no single-phase ducted unit in the catalogue, so the only honest single-unit answer is three-phase — which is now said out loud rather than assumed.

04 What is still not right

Limitations that remain after this round. None is a bug; all of them shape what the tool can tell you.

- **85 of 120 models have no manufacturer static pressure on file.** Braemar, Gree, Midea, Mitsubishi Electric, Mitsubishi Heavy and Samsung have none at all. Designs on those models cannot complete the static check until the figure is typed in from the data sheet. The full list is in `docs/static-pressure-audit.txt`.
- **58 of 120 models have no supplier cost**, so they cannot be costed or quoted without one being entered.
- **25 of 52 material lines are shipped placeholder rates**, not NAC prices. On cost-plus-fee they go straight through to the customer.
- **Plan reading by AI was never exercised here** — the AI endpoints are unreachable from this environment. The deterministic engines behind them were tested instead, which is the half that must be right every time.
- **Row-level security is unproven until it is applied.** Until then the sign-in screen is a door with no lock behind it.
- **There is one staff tier.** Any authenticated user is treated as NAC staff. No second tier was invented because none was asked for.
- **No load or performance testing** has been done at any point.

05 The five blockers, in order

Each one depends on the one before it. None needs a developer, and none takes long.

1 Apply the security policies

Run `designer/rls.sql` in the Supabase SQL editor. Until this is done the anon key that ships in the page source can still read every customer, every price and every design, whatever the sign-in screen says. The file explains each policy and carries its own undo block.

Proves: an unauthenticated visitor gets nothing. Customer signing pages keep working.

2 Prove the database really persists

Open `/db-selftest.html` on the deployed app, signed in, and run it. It creates, reads back, updates, lists and deletes real rows in `nac_settings`, `nac_quotes` and `nac_designs`, and checks a whole design survives with its rooms, equipment, materials, pricing and revisions. Test rows are prefixed ZZ-SELFTEST- and deleted afterwards. Then sign out and run it again — the results should differ, and that difference is step 1 working.

Turns P1 from NOT TESTABLE into a real answer. Do not skip the signed-out run.

3 Set the 25 material rates

HVAC Design Settings → Material rates. Run `node tools/material-price-report.mjs` for the list, or read it off the Materials tab where every placeholder is marked. Flexible duct, diffusers, dampers, grilles, cable, isolators and sundries are the bulk of it.

Until then every quote carries a confirmation naming the guessed dollars.

4 Create the customer and job records

Run `designer/crm-schema.sql` — additive, no drops, no renames — then open `/crm-migrate.html` and press **DRY RUN**. Read what it says it would do, especially the disagreements it found, before pressing APPLY. Nothing is written until you do, nothing you already have is overwritten, and running it twice changes nothing the second time.

Existing quotes, designs and the signing page are unaffected either way.

5 Connect ServiceM8

Set `SERVICEM8_API_KEY` in the Vercel project, then open `/api/servicem8-selftest`. It creates nothing — it reports whether the key is accepted, whether the records a job needs can be read, and whether failures can be recorded. Then accept one real quote and check the job appears.

Turns P8 from NOT TESTABLE into a real answer.

When all five are done, the answer at the top of this page changes on evidence rather than on optimism. Until then the honest position is that the application behaves correctly against everything that could be tested here, and four of the eight priorities depend on systems this environment was never able to reach.

06 How it was tested

WHAT	COUNT	WHAT IT ACTUALLY PROVES
Unit tests	388 in 24 files	The engines: plan reading, room measurement, loads, equipment, airflow, du zones, static pressure, materials, costing, warnings, customers, jobs, the two the PDF writer and the ServiceM8 handler.
Browser tests	16 suites	Real pages in Chromium with the database stubbed: persistence across reload failures, the pricing gate, the sign-in gate, the dialogs, the PDFs read back with customer and job chain, and the whole workflow from plan to signed quote.
Button audit	97 buttons	Every button on every tab has a real click handler, checked through the debugger than by reading <code>onclick</code> .
Page health	8 pages	No console errors and no failed requests other than the hosts this environment. Gated pages are reported as gated, not as blank.
Real plans	2	A dimension-chain builder plan and your printed-size brochure sheet, both ready to end.